



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

## SAFETY DATA SHEET

### Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

### 1. CHEMICAL IDENTIFICATION

#### Product Identification

Trade name: Terrazole

Product name: etridiazole 24% W/V EC

Product use: Fungicide

#### Relevant identified uses of the substance/mixture and uses advised against:

Used to control damping-off, root rot and stem rot caused by *Pythium* and *Phytophthora*. It is used on golf course turf, cotton and nursery ornamentals, non-bearing citrus, non-bearing coffee, and mixed with potting soil for greenhouse use. It is also used as a seed treatment on barley, beans, corn, cotton, peanuts, peas, sorghum, soybeans, safflower and wheat. Seven states hold special local need registrations for use in tobacco transplant floatbeds.

### 2. Hazard Identification

#### GHS-Classification

| Flammable liquids        | Category 4 |            |
|--------------------------|------------|------------|
| Acute toxicity           | Category 5 | Oral       |
| Acute toxicity           | Category 5 | Inhalation |
| Acute toxicity           | Category 5 | Dermal     |
| Carcinogenicity          | Category 2 |            |
| Aspiration hazard        | Category 1 |            |
| Acute aquatic toxicity   | Category 1 |            |
| Chronic aquatic toxicity | Category 1 |            |

#### GHS-Labeling





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#### Hazard statements

Danger:

Combustible liquid.

May be harmful if swallowed or in contact with skin.

May be fatal if swallowed and enters airways.

May be harmful if inhaled.

Suspected of causing cancer.

Very toxic to aquatic life with long lasting effects.

Precaution:

Obtain special instructions before use.

Do not handle until all safety precautions have been read and understood.

Keep away from heat/sparks/open flames/hot surfaces. - No smoking.

Avoid release to the environment.

Wear protective gloves/ protective clothing/ eye protection/ face protection.

IF SWALLOWED: Immediately call a POISON CENTER or doctor/ physician.

IF INHALED: Call a POISON CENTER or doctor/ physician if you feel unwell.

IF exposed or concerned: Get medical advice/ attention.

Do NOT induce vomiting.

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction.

Collect spillage.

Store in a well-ventilated place. Keep cool.

Store locked up.

Dispose of contents/ container to an approved waste disposal plant.

### **3. COMPOSITION/INFORMATION ON INGREDIENT**

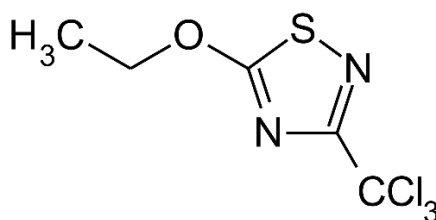
Chemical name: 5-ethoxy-3-trichloromethyl-1,2,4-thiadiazole [IUPAC]

Common name: etridiazole

Formula:  $C_5H_5Cl_3N_2OS$

Molecular weight: 247.518 g/mol

Structural formula:





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### Ingredients and composition:

| Component                                    | CAS No.    | Concentration       |
|----------------------------------------------|------------|---------------------|
| 5-ethoxy-3-trichloromethyl-1,2,4-thiadiazole | 2593-15-9  | $\geq 10 - < 30$    |
| Solvent naphtha (petroleum), heavy aromatic  | 64742-94-5 | $\geq 60 - \leq 80$ |
| butan-1-ol                                   | 71-36-3    | $< 1$               |

## 4. FIRST AID MEASURES

### Description of first aid measures

In case of swallowed:

DO NOT induce vomiting.

Rinse mouth with water.

Obtain medical attention.

In case of eyes contact:

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

If symptoms persist, call a physician.

In case of skin contact:

Remove contaminated clothing and shoes.

Wash off with warm water and soap.

If symptoms persist, call a physician.

In case of Inhaled contact:

Remove to fresh air.

Obtain medical attention.

## 5. FIRE FIGHTING MEASURES

Flash point: 69.3 °C (147.0 °F)

### Protective equipment and precautions for firefighters

Extinguishing Media:

Extinguishing media - large fires

Alcohol-resistant foam (on small fires)

Carbon dioxide (CO<sub>2</sub>)

Dry chemical

Unsuitable Extinguishing Media:

Water spray jet

Hazardous Combustion Products:

No hazardous decomposition products are known.



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### Further information:

Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

Prevent fire extinguishing water from contaminating surface water or the ground water system.

### Special protective equipment for fire-fighters:

In the event of fire, wear self-contained breathing apparatus.

Complete suit protecting against chemicals.

Full protective flameproof clothing.

## 6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

### Personal precautions:

Immediately evacuate personnel to safe areas.

Ensure adequate ventilation.

Wear suitable protective equipment.

### Environmental precautions:

Do not allow material to contaminate ground water system.

Do not allow uncontrolled discharge of product into the environment.

### Methods for containment /

### Methods for cleaning up:

Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13).

Dam up.

Large spills should be collected mechanically (remove by pumping) for disposal.

Observe government regulations.

## 7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

### Safe Handling:

Wear suitable protective equipment.

Keep container tightly closed.

Keep container tightly closed.



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Storage Conditions:

Keep in a dry, cool place.

## 8. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Personal Protective Equipment

Eyes Protection:

Face-shield

Skin and Body Protection:

Full protective suit

Respiratory Protection:

In the case of vapor formation use a respirator with an approved filter.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

|                      |                                 |
|----------------------|---------------------------------|
| Physical stage:      | Liquid                          |
| Color:               | Amber                           |
| Flash point:         | 63.9 °C                         |
| Density:             | 1.001 – 1.007 g/cm <sup>3</sup> |
| Solubility in water: | insoluble                       |

## 10. STABILITY AND REACTIVITY

Condition to avoid: Extremes of temperature and direct sunlight.

Materials to avoid: None known.

Hazardous decomposition products: No hazardous decomposition products are known.

Hazardous reactions: No dangerous reaction known under conditions of normal use.

## 11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, etridiazole:

|                             |               |
|-----------------------------|---------------|
| LD50 (Oral, rat):           | > 1,028 mg/kg |
| LD50 (Dermal, rabbit):      | > 5,000 mg/kg |
| LC50 (Inhalation, rat, 4h): | 5.7 mg/L      |
| Skin irritation (rabbit):   | Not irritant. |
| Eye irritation (rabbit):    | Not irritant. |



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Skin sensitization:

Causes sensitization in Guinea pig.

Classification: "May cause sensitization by skin contact."

## 12. ECOLOGICAL INFORMATION

### Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, etridiazole;

#### **Fish – Acute toxicity:**

Rainbow trout (*Oncorhynchus mykiss*),

LC50 (96 h): 2.4 mg/L

#### **Aquatic invertebrates - Acute toxicity:**

Daphnia magna (Water flea),

EC50 (48 h): 3.1 mg/L

#### **Algae – Acute toxicity:**

IC50 (72 h): 0.39 mg/L

## 13. DISPOSAL CONSIDERATIONS

Dispose of in accordance with local regulations.

## 14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

### ADR

UN No: 3082

Description of the goods: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
(Etridiazole)

Dangerous Goods Class: 9

Packaging Group: III

Classification Code: M6

Hazard identification No.: 90

Labels: 9

### IATA

UN No: 3082

Description of the goods: Environmentally hazardous substance, liquid, n.o.s.



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(Etridiazole)

|                        |                                                  |
|------------------------|--------------------------------------------------|
| Dangerous Goods Class: | 9                                                |
| Packaging Group:       | III                                              |
| Labels:                | 9                                                |
| Packing instruction:   | 914 (cargo aircraft)<br>914 (passenger aircraft) |

### IDMG

|                           |                                                                      |
|---------------------------|----------------------------------------------------------------------|
| UN No:                    | 3082                                                                 |
| Description of the goods: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,<br>N.O.S. (Etridiazole) |
| Dangerous Goods Class:    | 9                                                                    |
| Packaging Group:          | III                                                                  |
| Labels:                   | 9                                                                    |
| EmS Number 1:             | F-A                                                                  |
| EmS Number 2:             | S-F                                                                  |
| Marine pollutant:         | Yes                                                                  |

## 15. REGULATORY INFORMATION

Follow all regulations in your country.

## 16. OTHER INFORMATION

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.